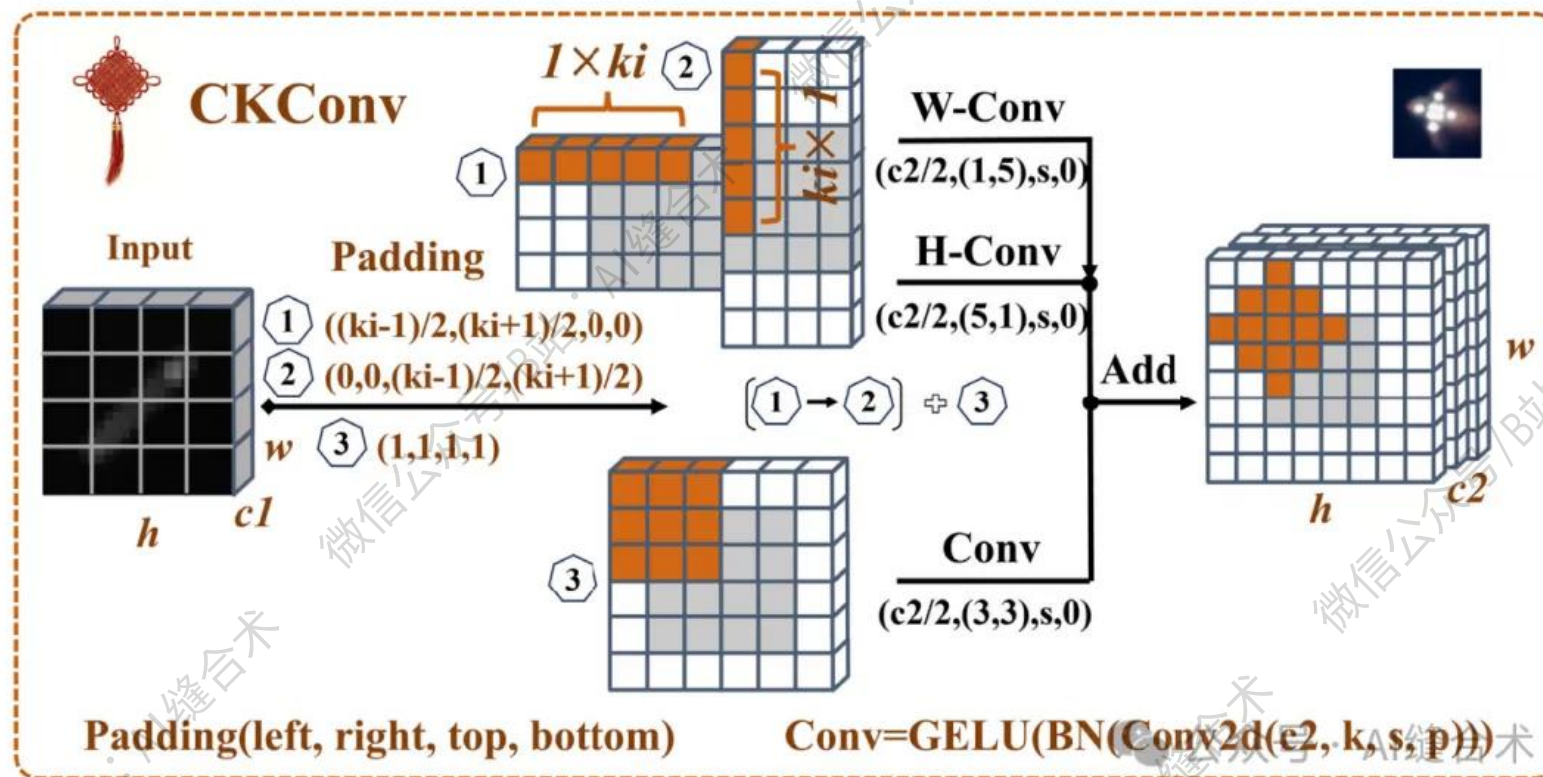


核心速览



论文题目: MSCK-Net: Multiscale Chinese Knot Convolutional Network for Dim and Small Infrared Ship Detection

中文题目: MSCK-Net: 用于弱小红外船舶检测的多尺度中国结卷积网络

论文链接: <https://ieeexplore.ieee.org/stamp/stamp.jsp?tp=&arnumber=11320952>

所属单位: 杭州电子科技大学, 汉阳大学等

核心速览:

针对红外弱小舰船目标（DSIRSTs）检测中数据集规模小、目标稀疏、背景复杂等问题，构建了包含 27138 张图像的 IRShip 数据集，提出了 CP-PB 离线数据增强和 Dense-O2O 在线增强策略，并设计了融合中国结卷积（CKConv）的多尺度网络 MSCK-Net，在多个数据集上实现了优越的检测性能。

<https://mp.weixin.qq.com/s/TpG4AHPgj6te6psdUOKFUw>

```
(conv): Conv2d(16, 16, kernel_size=(5, 1), stride=(1, 1), padding=(2, 0), groups=16, bias=False)
(bn): BatchNorm2d(16, eps=1e-05, momentum=0.1, affine=True, track_running_stats=True)
(act): GELU(approximate='none')
)
(k5_conv2): Conv2d(16, 32, kernel_size=(1, 1), stride=(1, 1), groups=16)
(k7_body): Conv(
  (conv): Conv2d(32, 16, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1), groups=16, bias=False)
  (bn): BatchNorm2d(16, eps=1e-05, momentum=0.1, affine=True, track_running_stats=True)
  (act): GELU(approximate='none')
)
(k7_head_h): Conv(
  (conv): Conv2d(32, 16, kernel_size=(1, 7), stride=(1, 1), padding=(0, 3), groups=16, bias=False)
  (bn): BatchNorm2d(16, eps=1e-05, momentum=0.1, affine=True, track_running_stats=True)
  (act): GELU(approximate='none')
)
(k7_head_v): Conv(
  (conv): Conv2d(16, 16, kernel_size=(7, 1), stride=(1, 1), padding=(3, 0), groups=16, bias=False)
  (bn): BatchNorm2d(16, eps=1e-05, momentum=0.1, affine=True, track_running_stats=True)
  (act): GELU(approximate='none')
)
(k7_conv2): Conv2d(16, 32, kernel_size=(1, 1), stride=(1, 1), groups=16)
)
(conv_fuse): Conv2d(96, 32, kernel_size=(1, 1), stride=(1, 1), groups=16)
)
input_tensor_shape : torch.Size([2, 32, 256, 256])
output_tensor_shape : torch.Size([2, 32, 256, 256])
```

哔哩哔哩/微信公众号：AI缝合术，独家整理！